The Price of Evaluation Choices in Portfolio Backtests*

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Abstract

A backtest reports a number, but that number depends on choices the reader never sees: which baseline was used, over which dates, at which trading cost, and at which random seed; we measure what those choices are worth. We first build a set of regime-aware, graph-based and neural portfolio methods inside one causal, cost-aware pipeline, and find that no method beats a simple baseline out of sample. That failure gives us a test bed with a known ground truth. None of the methods has an edge, so any performance still visible in these runs comes from how the results were reported. Across 345 manifest-backed runs on four fixed-instrument panels and two markets, we hold the data, allocator, cost model and execution rule fixed and change one reporting choice at a time. Ignoring trading costs adds 0.232 to the Sharpe ratio of a model that trades 56% of the portfolio a month, but only 0.010 to equal weight, so the effect grows with turnover. A benchmark measured over different dates adds 0.493 to the reported figure, and reporting a single favourite baseline rather than the full set adds 0.070 while reversing the sign of the conclusion. An uncorrected best-of-family p-value understates the corrected one by up to 0.321. We then pre-registered a prediction of failure and tested it on a sealed panel sharing no instrument with the development data. The prediction held, and the trading-cost result repeated out of sample.

CCS Concepts

• **Computing methodologies** → **Machine learning**; *Cross-validation*; • **Mathematics of computing** → *Time series analysis*; Regression analysis.

Keywords

portfolio optimisation, backtest overfitting, multiple testing, reproducibility, regime models, graph neural networks

1 Introduction

Many papers report strong risk-adjusted returns for regime-aware and graph-neural portfolio methods. We set out to reproduce that kind of result. To make the comparison fair, we built all of them inside one pipeline, so they see the same data at the same time and send their output through a shared allocator and backtester. No method can look better simply because it was given easier conditions.

Nine candidates failed that test: soft-regime conditional covariance, an uncertainty-aware hidden Markov risk model, a group-aware covariance variant, edge-masked graph attention, a training objective based on the information coefficient (IC), probability-gated expert heads, direct concatenation of regime probabilities, a VIX-conditioned regime model, and a dynamic graph-temporal transformer. Each was compared against conventional allocators on held-out data, and none cleared its pre-declared release gate (Table 2).

When the dynamic model is trained for the twenty epochs its configuration declares, it reproduces the published pattern, reaching a five-seed median Sharpe of 0.930 on the development window against 0.792 for equal weight. The same runs evaluated on held-out data give 0.604 against 0.718. Section 5 traces how much of the development-window advantage each reporting choice supplied.

These failures give the rest of the paper a fixed reference point. None of the methods has an edge, so the true value of any advantage a backtest of them reports is zero, and an advantage that still appears in a table was supplied by the evaluation rather than by a method. Reporting choices can therefore be measured directly. Holding the panel, the strategy, the allocator, the cost model and the execution rule fixed, we change one reporting choice at a time and record what happens to the reported number. We call that change *protocol inflation*.

Our contributions are:

(1) An audited benchmark of 345 manifest-backed causal runs across four fixed-instrument panels and two markets, under one shared allocator, cost model and release gate.
(2) A measurement of protocol inflation. We price six common reporting choices across fourteen single-choice cells of Table 3, changing one thing at a time and pairing each with a control where one exists, plus a fifteenth cell that compounds three of them.
(3) A confirmation test on a sealed panel that shares no instrument with the development data. We recorded our prediction, which was that the methods would fail, before opening it.
(4) A reproducibility harness that blocks rather than guesses when something is missing, and a demonstration that the whole confirmation cycle re-runs to identical numbers.

Our claims are about measurement rather than investment. Each number in this paper prices an evaluation choice, and that price is a property of the protocol rather than of any method's merit. The paper accordingly contains no investable result. The compound cell of Table 3, which reports a Sharpe of 1.095, adjusts to 0.604 and trails equal weight's 0.718 over the same dates.

2 Related Work

This paper builds on the literature on data snooping and backtest overfitting. White's Reality Check [23] and Hansen's test for superior predictive ability [10] correct a reported edge for the size of the search that produced it. The deflated Sharpe ratio [4] and the analysis of backtest overfitting [3] bound how much a reported Sharpe should be discounted given the number of trials. Harvey,

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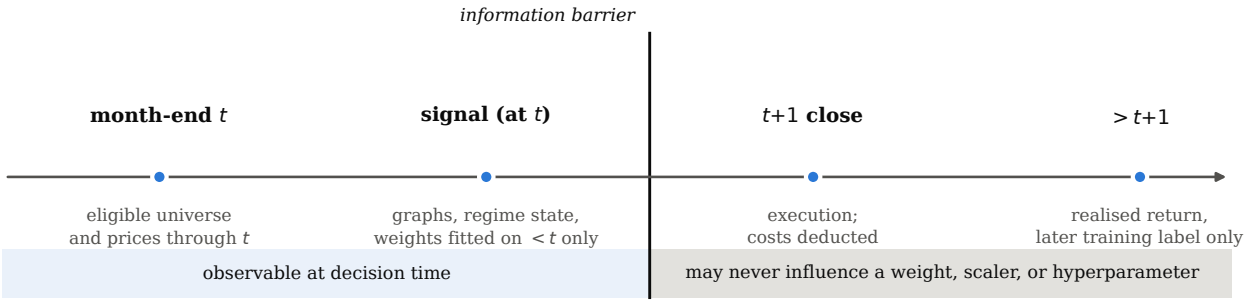


Figure 1: The causal decision timeline. At month-end t the pipeline sees only data through t; graphs, regime states, scalers and weights are fitted strictly before t; execution happens at the following close with costs deducted; the realised return may enter only as a later training label. No scaler, graph edge, regime feature, hyperparameter, membership record or normalisation statistic may cross the barrier.

Liu and Zhu [12] make the same argument for the cross-section of expected returns, and Harvey and Liu [11] for backtesting generally. Arnott, Harvey and Markowitz [2] propose a protocol for machine-learning backtests.

Most of these results are theoretical, showing that a reported figure deserves a discount and bounding the size of that discount for a given number of trials. We contribute the empirical counterpart by pricing each individual choice on runs where the true edge of every method is known to be zero. The two quantities are complementary rather than competing, since a deflated Sharpe tells a reader how much to discount a number while our measurements identify which decision produced it.

Kapoor and Narayanan [14] document leakage as a reproducibility problem across machine-learning-based science and argue for treating it as a first-class object of study. We adopt that stance in a portfolio setting, where leakage is one of several conventions that inflate a result, and by our measurements not the largest.

Two studies published in 2026 address closely related questions. Zhang et al. [25] construct a one-switch benchmark that isolates decision-time leakage and measures the alpha it manufactures. Our treatment of the development-window choice asks a similar question, and we agree with them that varying one thing at a time is the right experimental form. The differences are in scope and in the runs being read. Their study isolates leakage, while we treat it as one of six choices and find its effect uneven. The same boundary choice is worth +0.090 at our frozen training budget and +0.326 at the declared one, so leakage is among the smaller effects in its headline instance without being small in general. A leakage benchmark also has to work out how much of an observed edge was real. Our runs have no edge to divide up, since every method has already failed on held-out data, so anything recovered from them is an artefact by construction. Yin et al. [24] quantify implementation risk, the divergence between backtested and deployable performance, which is the mechanism underlying our cost row. We add a measurement of how that distortion varies between methods. It grows with turnover, and a low-turnover control makes that growth visible.

On the methods side, DeMiguel, Garlappi and Uppal [6] established that naive diversification is hard to beat. Our ladder relies on that result and complicates it, since the identity of the winning baseline changes from one panel to the next. The allocator ladder draws on hierarchical risk parity [18], shrinkage covariance [17] and conditional value-at-risk optimisation [22]; the regime models follow Hamilton [9] and Ang and Bekaert [1]. The candidates we implement follow recent work on graph-attention portfolios [15], multivariate hidden Markov selection [19], the implementable efficient frontier [13], forecast stabilisation [5] and machine-learning asset pricing [8].

3 Experimental Apparatus

3.1 Causal timeline

Every method shares the decision timeline drawn in Figure 1. A decision made at month-end t may use prices and membership only through t. Graphs, regime states, scalers, model weights and hyperparameters are fitted on strictly earlier data, and the realised return becomes visible only afterwards, as a training label. The barrier between what a decision may and may not see is enforced in code rather than left to convention, so no look-ahead path exists. Figure 2 then traces the path a run must travel before it may support a claim. Every stage on that path can block the run, and no stage can improve it.

3.2 Data and protocol

The panels are fixed lists of exchange-traded funds, declared before any price was retrieved: 30 US-listed funds from 2010, a wider 50-fund US panel, 12 Indian funds from 2014, and a second, disjoint list of 30 US-listed funds reserved as the sealed panel for the confirmation test of Section 6. Prices are daily adjusted closes retrieved through the `yfinance` interface, with the retrieval timestamp, library version and response hash recorded in each run manifest. Decisions are made at each month end, executed at the next close, and charged 10 basis points one way, with a 10% cap on any single

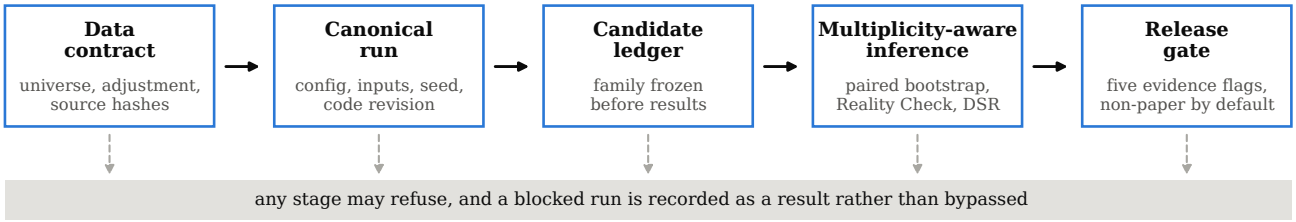


Figure 2: The release path from raw data to a claim. Each stage can only refuse a run; nothing on the path improves one, so the path acts as a sequence of gates rather than a processing chain. A blocked run is recorded as a result rather than bypassed, so a failed candidate remains visible in the record rather than quietly disappearing.

position. Rebalancing is monthly except in the frequency robustness cells of Table 5, which use quarterly decisions.

3.3 Metrics

Sharpe ratios are computed from daily returns net of transaction costs and annualised by $\sqrt{252}$, with the risk-free rate set to zero throughout. Our claims concern two readings of the same runs, for which a common rate offset is second order. Maximum drawdown is taken on the daily net asset value. Conditional value-at-risk (CVaR) is the mean of the worst 5% of daily net returns. Turnover is one-way and reported per executed rebalance. Every comparison between two strategies uses only the dates both of them realise.

3.4 Shared allocator

Every method, conventional or learned, produces either a cross-sectional score or a covariance matrix. A covariance matrix is allocated by hierarchical risk parity [18], which clusters assets on the correlation distance $\sqrt{(1-\rho)/2}$ and splits capital recursively by inverse cluster variance. A score is allocated by a long-only mean-CVaR programme [22] that maximises the score, taken as the expected-return estimate, less a 5% conditional value-at-risk penalty and a turnover penalty on the distance from the current holding. Both paths are fully invested and capped at 10% per asset, and both are executed by the same cost-aware backtester, so a learned method cannot win by being given an easier execution model.

3.5 Candidates and baselines

A method is a candidate if it fits a latent state or trains parameters by gradient descent, and a baseline if it is a deterministic function of observable returns. This split fixes which methods enter a family-wise correction, so it was frozen in version control before any confirmation result was read.

3.6 Provenance

Each panel is governed by a versioned contract declaring its universe, adjustment convention, provider metadata, coverage requirements and source hashes. Validation precedes every run, and a stale report is rejected rather than reused.

A canonical run records its configuration, input hashes, code revision, targets, executions, holdings, daily net asset value, diagnostics and seed, and every number in this paper resolves to one of these records.

3.7 Release gate

Runs are classified non-paper by default, promotion requires five separate evidence flags, and the summariser will not export a run that has no classification. Every run in this study remains unpromoted, which is the state the design intends. Our claims concern reporting conventions rather than performance, and the gate keeps the two apart by construction, since nothing in this study can be exported as a performance result.

4 Candidate Methods

Every candidate was evaluated against a ladder of conventional allocators on matched dates, using a paired 21-session moving-block bootstrap with 5,000 replications [16, 21]. The choice of block length is not doing the work, since repeating the closest confirmation comparison at 10, 21, 42 and 63 sessions returns p between 0.131 and 0.146. Table 1 specifies each candidate family, its matched comparator, and what that comparison isolates; Table 2 lists each test and the p-value it returned. No entry approaches conventional significance, and the pattern holds across regime conditioning, graph construction, training objective and architecture alike. Risk adjustment does not change the picture either, since on the US panels no strategy earns a statistically significant positive five-factor alpha [7] under Newey–West standard errors [20]. We do not run the equivalent regression on the Indian panel, since the five factors are built for the US cross-section.

A final check rules out undertraining. Our frozen configuration trains the neural candidate for five epochs, although its declared

Table 1: Candidate specifications: mechanism, matched comparator, and what each comparison isolates. Every candidate sends its output through the shared allocator of Section 3, so a comparison differs only in the component named. Seeds are the prescribed set $\{7, 17, 29, 43, 59\}$ where a method is stochastic.

Family	Mechanism	Matched comparator	Isolates	Seeds
Soft-regime cond. covariance	expanding soft k-means, 3 states, on four market features; state-mixed shrinkage covariance, 756-session lookback	matched unconditional risk: the same HRP allocation on an unconditional covariance	isolates the state mixture	1
HMM cond. risk (2, 252)	expanding Gaussian HMM, 2 states, 252-session covariance lookback, entropy fallback to unconditional risk	matched HRP-756, the stronger unconditional control; the lookback-matched HRP-252 gives p=0.209	isolates state conditioning	1
9-cell HMM grid	the same model over states $\in \{2, 3, 4\}$ and lookback $\in \{63, 126, 252\}$ sessions	matched HRP-756	prices the search over those two axes	1
Group-aware cond. covariance	as the HMM candidate, but state conditioning retained only within five ex-ante asset groups	matched global-HMM risk	isolates cross-group conditioning	1
Edge-masked graph attention	attention over the sparsified graph in place of similarity message passing	similarity passing: the same encoder with fixed weights	isolates learned edge weighting	5
IC training objective	ranking-aligned loss on the cross-sectional information coefficient	likelihood-only: the same model under a Gaussian likelihood	isolates the training objective	5
Regime-conditioned variants	regime probabilities entering the model by direct concatenation, by gated expert heads, or through a VIX covariate	regime-free model: the same architecture without state input	isolates regime conditioning	5
Dynamic graph-temporal	partial-correlation graph, 252-session lookback, 5% edges; 2 graph layers into a 2-layer, 4-head Transformer, d=48, sequence length 12	its graph-only and temporal-only ablations, and equal weight	isolates each stage, then the whole model	5

Table 2: Every proposed candidate, its matched comparator, and the one-sided paired moving-block p-value for a positive advantage. The final three rows are the same model, which beats both of its own ablations and still loses to equal weight. Thirteen tests cover nine candidates. The information-coefficient objective appears on two panels, the HMM entry appears with and without its search correction, and the dynamic model is tested against two ablations and one baseline. All rows resolve to manifest-backed runs (Section 3).

Candidate	Compared against	p
Soft-regime cond. covariance	matched uncond. risk	0.909
HMM cond. risk (2, 252)	matched HRP-756	0.359
same, corrected for the 9-cell search	matched HRP-756	0.680
Group-aware cond. covariance	matched global-HMM risk	0.732
Edge-masked graph attention	similarity passing	0.474
IC training objective (US)	likelihood-only	0.298
IC training objective (India)	likelihood-only	0.998
Probability-gated regime experts	regime-free model	0.719
Direct probability concat.	regime-free model	0.519
VIX-conditioned regime model	regime-free model	0.717
Dynamic graph-temporal, full	own graph-only ablation	0.007
same	own temporal-only abl.	0.023
same	equal weight	0.705

configuration specifies twenty, so the failures above could in principle reflect the budget rather than the architecture. We therefore ran the declared budget on the development panel across the same five seeds. At twenty epochs the development median rises from 0.763 to 0.930, moving from below equal weight (0.792) to 0.137 above it. On held-out data the same runs give 0.604 against equal weight's 0.718, which is 0.115 below. Quadrupling the training budget therefore adds 0.167 Sharpe on the development window and adds nothing out of sample. Differences in this paragraph, as in Table 3, are computed from unrounded values. That 0.326 difference appears as its own row in Table 3, and the spread across seeds widens from 0.185 to 0.321. All other results in this paper use the five-epoch setting, the configuration that entered the confirmation test.

The edge-masked attention model and the graph-temporal transformer are the graph-neural candidates named in the introduction; both learn by passing messages over the sparsified partial-correlation graph specified in Table 1. The last three rows of Table 2 describe the graph-temporal model and are worth examining separately. The full graph-temporal model beats its own graph-only (p=0.007) and temporal-only (p=0.023) ablations, and then loses to equal weight (p=0.705). Each component of the model contributes something, and the model as a whole is still worse than the simplest rule available. An ablation study cannot detect this by itself, since every comparison it makes is against a degraded version of the same model.

5 Protocol Inflation

Table 3 and Figure 3 give the main result. Each row reads the same runs twice. The Reported column applies one favourable evaluation choice, and the Adjusted column removes it. Nothing else changes between the two columns, so the gap between them is the price of that single choice.

5.1 Trading costs

Leaving costs out adds 0.232 to the Sharpe ratio of a model that trades 56% of the portfolio a month. The same omission adds only 0.010 to equal weight, which trades 2.6%. The effect therefore depends on how much a strategy trades. In most papers the method that trades most is the one being put forward, so the omission helps the proposal more than the baseline. The same pair reproduces on the sealed panel at 0.167 against 0.007. The measured 0.232 exceeds the direct cost drag, since 56% one-way monthly turnover at 10 basis points removes roughly 1.3 percentage points of annual return; the remainder of the Sharpe difference comes from compounding on a lower daily net asset value and from the volatility denominator.

Table 3: Measured inflation from each evaluation choice, grouped by the choice isolated. Each row reads the same runs twice, holding the panel, strategy, allocator, cost model and execution rule fixed. Rows marked *sealed* were computed on the confirmation panel, which had never been examined when the earlier rows were produced. Inflation is the reported value minus the adjusted one, computed from unrounded values; columns are rounded to three decimals, so a difference taken from them can disagree in the last digit. The search-multiplicity group reports p-values rather than Sharpe ratios, so its inflation is a difference in p. Sharpe cells aggregate seeds by the estimator the frozen protocol declared, mean at five epochs and median in the declared-budget experiment.

Choice	Panel / strategy	Reading	Reported	Adjusted	Inflation
Development boundary	US 30-ETF: Causal dynamic graph-temporal model, 5 epochs	development vs held out	0.775	0.685	**+0.090**
	US 30-ETF, declared 20-epoch training budget: Dynamic graph-temporal, 5 seeds (equal weight: 0.792 development, 0.718 outer)	development vs held out	0.930	0.604	**+0.326**
Seed selection	US 30-ETF: Causal dynamic graph-temporal model, outer window	best seed vs median	0.734	0.729	**+0.005**
	US 50-ETF: Causal dynamic graph-temporal model, development window	best seed vs median	1.261	0.759	**+0.502**
	Confirmation panel, 2020+ secondary window (sealed): Dynamic graph-temporal; best baseline on this window is 0.737	best seed vs median	0.799	0.714	**+0.085**
Trading costs	US 30-ETF: Causal dynamic graph-temporal model (mean one-way turnover 56%/month)	gross vs net of 10 bps	0.731	0.500	**+0.232**
	US 30-ETF: Equal weight (low-turnover control: the shortcut should barely move it)	gross vs net of 10 bps	0.720	0.710	**+0.010**
	Confirmation panel (US 30-ETF, sealed): Dynamic graph-temporal, seed 7 (48.9%/month turnover)	gross vs net of 10 bps	0.735	0.568	**+0.167**
	Confirmation panel (US 30-ETF, sealed): Equal weight (low-turnover control: 2.0%/month)	gross vs net of 10 bps	0.680	0.673	**+0.007**
Benchmark choice	Confirmation panel (US 30-ETF, sealed): Dynamic graph-temporal, best seed vs the six pre-declared baselines	one baseline vs ladder	+0.045	−0.025	**+0.070** †
Date alignment	Confirmation panel, quarterly rebalance (sealed): Dynamic graph-temporal seed 7 (24-quarter warm-up starts 2020) vs equal weight	own dates vs matched	+0.538	+0.045	**+0.493**
Search multiplicity	US 30-ETF: HMM cond. risk (2, 252), 9-cell grid vs matched HRP-756 control	best cell vs corrected	0.359	0.680	**+0.321**
	Confirmation panel (US 30-ETF, sealed): Seven-candidate confirmation family vs strongest baseline	best cell vs corrected	0.131	0.344	**+0.213**
	Confirmation panel, 2020+ secondary window (sealed): Seed 7 alone vs the seven-candidate family, against the strongest baseline	best cell vs corrected	0.065	0.185	**+0.120**
Combined choices	US 30-ETF, declared 20-epoch budget: Development window, best seed, zero costs, against held-out median at 10 bps	three choices at once vs adjusted	1.095	0.604	**+0.491**

† the sign of the conclusion changes, not merely its magnitude.

5.2 Benchmark choice

The choice of comparator can flip the sign of a conclusion. On the development panel the strongest simple baseline was equal weight, and a study that trusted its development evidence would adopt equal weight as the comparator. On the confirmation panel that choice decides the sign of the result. The best confirmation seed beats equal weight by 0.045 Sharpe yet finishes 0.025 behind the strongest baseline on that panel, so the same run supports either conclusion depending on the comparator it is shown against. The model, the data, the dates and the costs are identical in the two comparisons; only the benchmark differs, and it was selected before any model was trained. The claim can therefore be verified only when the full set of baselines is shown.

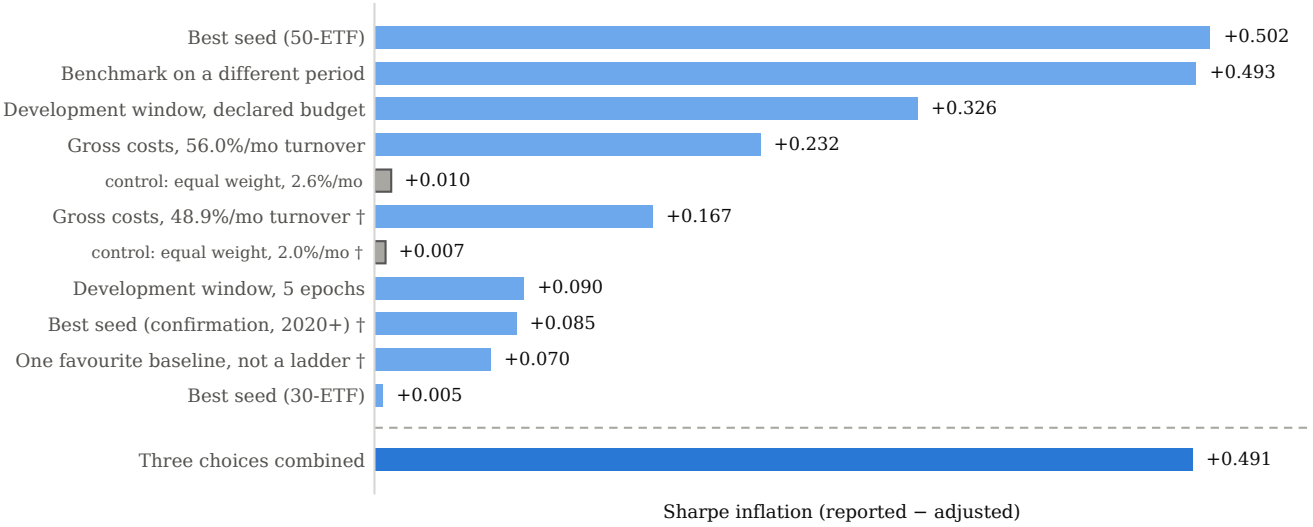


Figure 3: Measured inflation per evaluation choice. Grey bars are low-turnover controls, drawn directly beneath the high-turnover cell they pair with; the same choice barely moves them, and that contrast is the evidence that the convention is not neutral. The dashed rule separates the compound reading, drawn in a deeper blue because it combines three choices rather than pricing a single one. Rows marked † come from the confirmation panel, which stayed sealed until every configuration was frozen (Section 6).

5.3 Mismatched dates

This is the largest effect we measure and the hardest to notice. A model that needs a long warm-up starts later than its baselines. Under quarterly rebalancing our dynamic model needs 24 completed quarters before its first decision, so it starts in 2020 while the baselines start in 2014. A table that lists both without noting the dates shows 1.185 for the model against 0.646 for equal weight, an apparent +0.54 edge that would be the strongest result in this paper. Re-running every baseline over the model's own dates reduces the edge over that same benchmark to +0.045, and removes the lead altogether, since HRP (252) on matched dates sits within 0.017 of it (Table 6). Each individual run is correct and reproducible; the error lies only in placing two of them in the same column, and it stays hidden unless observation counts are printed beside the results.

5.4 Combined choices

Applying the development window, the best seed and zero costs to the same run reports a Sharpe of 1.095, against 0.604 when the same model is read on held-out data at the median seed and full costs. The three choices compose to +0.491, and because each is applied in sequence to one run the total is the sum of the sequential steps rather than evidence of interaction. The reported value is the point of the exercise. A model with no held-out edge, read under three ordinary conventions, reaches a Sharpe of 1.095.

5.5 Seed choice

Reporting the best seed instead of the median is worth 0.005 on one panel and 0.502 on another, using the same model, allocator and data. How many seeds are needed before a single-seed result misleads therefore depends on the panel, and cannot be carried over from another study.

Table 4: Confirmation-panel ladder. Net of 10 bps one-way costs, 10% cap, one-session execution lag, monthly rebalance, first signal 2014-01-31. Turnover is mean realised one-way turnover per execution. Italicised rows are the pre-registered candidates. No candidate leads, nor does any buy rank by taking more drawdown, tail, or trading risk.

Strategy	Sharpe	Max DD	$CVaR_{95}$	Turn.
Partial-corr. hub-avoid, CVaR	**0.748**	−34.3%	−2.29%	14.4%
PCC hub-avoid, CVaR	0.723	−36.0%	−2.32%	10.9%
Inverse volatility	0.708	−38.3%	−2.57%	2.2%
HRP (252)	0.704	−37.7%	−2.48%	7.6%
HMM cond. risk (2, 252)	0.697	−37.7%	−2.48%	8.7%
HRP (756)	0.695	−38.1%	−2.51%	5.6%
Soft-regime cond. covariance	0.686	−37.7%	−2.50%	8.1%
Equal weight	0.678	−39.2%	−2.66%	2.0%
Dynamic, 5-seed median	0.681	—	—	48.9%
seed range	0.598–0.723	—	—	—

6 Pre-Registered Confirmation

We reserved a confirmation panel of 30 US-listed ETFs sharing no traded ticker with the development universe. Before opening it we committed a hypothesis, the exact set of runs permitted, the decision statistic and a stop rule to version control, and we predicted that the candidates would fail. Recording the direction of the prediction in advance is what makes the outcome informative. A confirmed expectation carries little weight when the expectation was never written down.

Table 5: Cap and rebalance-frequency robustness (net Sharpe). A 2% cap is arithmetically impossible on a 30-asset universe and is not attempted. Both regime candidates fail outright under quarterly rebalancing.

Strategy	10% mon.	5% mon.	10% qtr.
Partial-corr., CVaR	0.748	0.728	0.672
PCC hub-avoid, CVaR	0.723	0.722	0.667
Inverse volatility	0.708	0.708	0.674
HRP (252)	0.704	0.698	0.668
HRP (756)	0.695	0.699	0.665
HMM cond. risk	0.697	0.696	**fails**
Soft-regime cond. covariance	0.686	0.690	**fails**
Equal weight	0.678	0.678	0.646

The prediction held, and Table 4 shows the outcome, with no candidate leading the ladder. We ran fourteen paired comparisons, seven candidates (the two regime methods and the dynamic model at its five prescribed seeds) each tested against two comparators. For multiplicity, each prescribed seed of the dynamic model counts as its own trial. The smallest p-value among them is 0.131. Table 4 reports the five seed runs as a median and a range rather than as separate rows. The three architectures carried into confirmation span the distinct mechanisms among the nine development candidates, namely regime-conditioned risk estimation, soft-regime allocation, and a learned dynamic model; the other six are variants within those three families. Correcting for the whole family, the Reality Check gives 0.718 against equal weight and 0.344 against the strongest baseline. The deflated Sharpe ratio, the probability that the best observed Sharpe exceeds zero given thirteen trials [4], is 0.484. The best strategy on that panel is a baseline rather than a candidate, and it performs about as well as thirteen skill-free trials would be expected to produce.

6.1 Baseline ranking

Equal weight was strongest on the development panel, with the graph and CVaR baselines behind it, and that order reverses on the confirmation panel, so no single baseline leads everywhere. Neither universe was selected with this reversal in mind. A study that commits to one favourite baseline can therefore obtain a win or a loss from that choice alone.

6.2 A near miss

On the declared secondary window one seed leads the table at 0.799 against the strongest baseline's 0.737, with p=0.065, a value close enough to conventional significance to anchor a headline claim. The apparent edge survives none of the four stricter readings of the same runs. The five-seed median is 0.714 and the worst prescribed seed on this window is 0.597, the first below the best baseline and the second below all six. The family-wise Reality Check returns 0.185, and the deflated Sharpe returns 0.513. The seed was not promoted, and we report the episode as a pre-registered instance of the paper's argument, with the gap between an apparent finding and no effect coming down entirely to which reading was applied.

Table 6: The quarterly cell, read two ways. The model's 24-quarter warm-up starts it in 2020 while the baselines start in 2014; every run is individually correct. Re-running the baselines over the model's own window collapses a +0.54 edge to +0.045 against equal weight and +0.017 against the strongest matched baseline. The model realises 17 fewer sessions than the baselines because its first execution follows its first signal by one session and its warm-up completes intra-quarter.

Strategy (quarterly)	Sharpe	Obs.
As printed: benchmarks from 2014, model from 2020		
Dynamic, seed 7	1.185	1569
Equal weight	0.646	3097
Date-matched: all runs from 2020-03-31		
Dynamic, seed 7	**1.185**	1569
HRP (252)	1.168	1586
Inverse volatility	1.162	1586
Equal weight	1.140	1586
PCC hub-avoid, CVaR	1.139	1586
Partial-corr., CVaR	1.135	1586
HRP (756)	1.127	1586

6.3 Robustness

Table 5 varies the portfolio cap and the rebalance frequency. Two of the declared cells could not be run, one for a trivial reason and one for an informative one. A 2% cap is infeasible on a 30-asset universe, since $30 \times 0.02 = 0.60$ leaves no way to hold a fully invested long-only portfolio. The more informative failure is that both regime candidates break outright under quarterly rebalancing. Each one estimates the market state on the rebalance clock, so quarterly decisions cut the number of state observations by two thirds while the warm-up requirement stays the same. Every conventional allocator handles the same change without any modification. Fragility of this kind is a separate defect from weak performance, and it stays invisible in any study that reports only its preferred monthly setting.

7 Limitations

Our panels are fixed-instrument ETF universes rather than point-in-time index constituent histories, so we make no claim about index replication or capacity. Both come from the same provider and overlap in calendar time, which makes the confirmation an independent set of instruments but not an independent data source. Declaring the lists before retrieval does not remove survivorship, since a fund had to exist to be named, but both readings of any row share the same universe, so the effect largely cancels in the difference we report. The magnitudes belong to these panels, strategies and windows. They show that the inflation is measurable and large enough to matter, but they are not a correction factor to subtract from anyone else's results. The Indian panel used here is a fixed list of exchange-traded funds with fully processed prices. A separate effort on Indian single-stock data, covering corporate actions and identity mapping, is still in progress and contributes nothing to the claims in this paper. The cap and frequency variants of the neural model were run at a single seed.

8 Reproducibility

Every number in this paper resolves to a manifest-backed run that records its inputs, hashes, code revision and seed, and a single committed script drives the whole confirmation cycle. Re-executing that script at a later code revision, after a refactor that left the effective evaluation windows unchanged, reproduced all 26 runs bitwise. Daily returns were identical for the ten stochastic model fits as well as for the deterministic ones. The contracts, ledgers, amendments, run manifests and exact commands needed to repeat the study will be made public when the paper is published.

9 Conclusion

This study set out to reproduce the reported outperformance of regime-aware and graph-neural portfolio methods. Nine candidates, built inside one causal, cost-aware pipeline, all failed against simple baselines on held-out data, and a prediction of that failure, registered before a sealed confirmation panel was opened, held. The failed runs then became the measuring instrument. With the true edge known to be zero, the gap between a favourable reading and an adjusted one prices the evaluation choice that produced it, and the prices are material. A benchmark from different dates adds 0.493 Sharpe, the best of five seeds adds up to 0.502, and one run read under three ordinary choices at once reports 1.095 where the adjusted figure is 0.604. At its declared training budget the dynamic model reproduces the published pattern on the development window, 0.930 against equal weight's 0.792, and gives all of it back out of sample. None of these effects requires an incorrect run, and every one disappears when only the preferred configuration is shown.

Each of these prices attaches to a reporting decision that every backtest study makes. A study should report the full ladder of simple baselines rather than a favourite, since the leading rule changes with the panel and fixing one is worth +0.070 here while reversing the sign of the conclusion. Every comparison should carry the number of observations on each of its sides, since a warm-up gap that silently shifts one start date by six years adds +0.493 without any run being wrong. Seeds should be reported as a distribution rather than as a maximum, a choice worth up to +0.502 and by an amount that varies from panel to panel.

Costs belong in the returns and turnover beside them, since omitting costs adds +0.232 to a high-turnover method and +0.010 to its low-turnover control, a contrast that repeated on the sealed panel at +0.167 against +0.007. Inference should price the search rather than the survivor, as an uncorrected best-of-family p of 0.359 becomes 0.680 once the other eight cells of the same grid are admitted. The boundary between development and held-out data should be stated wherever results cross it, a choice worth +0.090 at our frozen training budget and +0.326 at the declared one.

None of our own results survives these adjustments, and the pre-registered prediction said none would. An evaluation whose true edge is known to be zero is precisely what made each of these prices measurable.

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